Disaster Response Unmanned Aerial Vehicle (UAV) RFP

Background

This year has seen one of the largest fire seasons, in the west, on record. These fires covered a large portion of the west coast including California, Oregon and Washington, leaving a trail of destruction, destroying forests, homes and whole towns. Many people had to scramble to get of the way and some even lost their lives. Gov. Jay Inslee of Washington State referred to these as "… climate fires …" resulting from the change in the climate that we are experiencing. The expectation is that, with climate change, these fire seasons will get worse, driven by increasing hotter and dryer summers, with wetter winters producing more fuel.

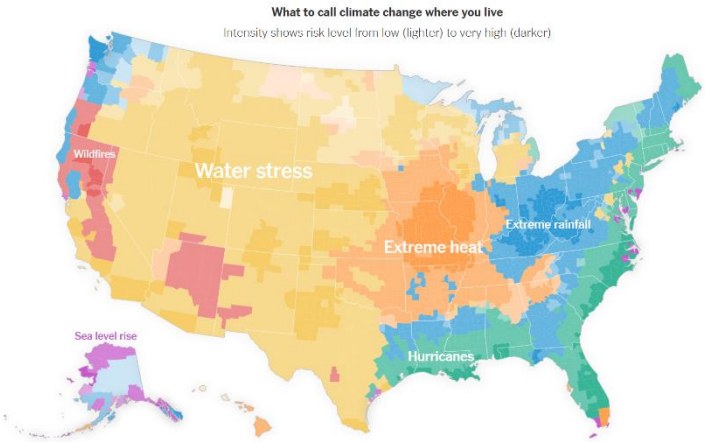

Figure 1 From New York Times (sept.18, 2020)

In addition, the south east of the United States is experiencing annual hurricanes, which are being described as more powerful and more frequent every year. They bring high winds, rain and flooding to large areas.

These types of climate disasters are wide-spread, covering large amounts of territory where emergency response may be hampered by lack of situational awareness, unreliable, damaged or just no communications, making response difficult, potentially endangering many lives.

In order to help coordinate a response, provide the necessary situational awareness and maintain open and reliable communications to a large disaster area, a rapidly deployable fleet of autonomous unmanned aerial vehicles (aUAV) is to be used. Thus, the proposal here is to design a medium to high altitude, high endurance UAV, carrying a surveillance payload to provide the necessary information (payload is defined below).

The fleet of UAVs will communicate with each other and allow coordinated coverage of a large area, while allowing units to swap out when they need refueling. This operational scheme is provided to give context to the project, however, the design of such an autonomous operations scheme in the current US airspace is beyond the scope of the project.

Requirements (M) = Mandatory Requirement (T) = Tradable requirement

- **All requirements can be questioned if data from completive airplanes or market data can be shared to provide evidence for the change.**
- **General Requirements for an autonomous UAV airplane**
  - Entry Into Service (EIS)
    - (M) 2022 for first deployment of the UAV
  - Payload
    - (M) High resolution visual cameras
    - (M) Infra-Red and Thermal cameras
    - (T) Night vision capable camera
    - (M) Mobile cell phone satellite (ie flying cell tower)

        - (M) Atmospheric sensors to provide weather and air quality data
        - (M) ADS-B transponder
        - (M) Autonomous aircraft control system
    - Design Endurance mission
        - (M) 12-hour duration
        - (T) 18-hour duration
    - Coverage Area
        - (M) 25 mile wide by 250 mile long observation / communication coverage zone
        - (T) 50 mile wide by 500 mile long observation / communication coverage zone
    - Design Speed
        - (M) 120kt cruise speed
        - (T) 250kt dash speed
    - Performance
        - Takeoff / Landing
            - (M) Short field takeoff / landing < 1,000ft
            - (T) Vertical takeoff / landing
        - Operating Altitude
            - (T) Altitude required to meet cell coverage and camera observation zone, not to exceed 31,000ft
    - Certification
        - (M) Capable of VFR and IFR flight with a remote and autopilot
        - (T) Capable of full autonomous flight within US commercial airspace
        - (M) Meets applicable certification rules in FAA 14 CFR Part 23 and FAA 14 CFR Part 107
            - Since UAV regulations may not fully cover operation of this vehicle, assume that it needs to meet the necessary requirements to operate in US commercial airspace, as well as meet the airworthiness requirements of general aircraft.
        - (T) Provide systems and avionics architecture that will enable full autonomous operations

Design Objectives

- Determine the payload weight for the required for equipment.
- Determine the aircraft size and propulsion system to meet the required mission.
- Optimize the structural arrangement to allow for easy transportation and rapid deployment in the disaster zone.
- Minimize production cost by choosing materials and manufacturing methods appropriate for a production rate that supports building a 100 aircraft per year.

Other features and considerations

- Flying qualities should meet CFR Part 23 & CFR Part 107.
- Identify all systems functionality and components that are required for the aircraft to operate in both controlled and uncontrolled airspace.
- List the equipment required.
- Consider what features will be basic and which will be optional, ie. specific equipment depending on what type of disaster the aircraft will be responding to.

Report and Design Data Requirements

The technical report shall present the design of this aircraft clearly and concisely; it shall cover all relevant aspects, features, and disciplines. Pertinent analyses and studies supporting design choices shall be documented.

Full descriptions of the aircraft are expected along with performance capabilities and operational limits. These include, at a minimum:

1. A description of the design missions defined for the proposed concepts for use in calculations of mission performance as per design objectives. This includes the selection of cruise altitude(s) and cruise speeds supported by pertinent trade analyses and discussion.
2. Aircraft thumbprint sizing charts along with the design decision rationale as to why the major components, such as wing area, aspect ratio, thrust/power, etc, were selected.
3. Aircraft performance summaries shall be documented and the aircraft flight envelope shall be shown graphically.
4. Range / Endurance chart(s)
5. A V-n diagram for the aircraft with identification of necessary aircraft velocities and design load factors.
6. Materials selection for main structural groups and general structural design, including layout of primary airframe structure as well as the strength capability of the structure and how that compares to what is required at the ultimate load limits of the aircraft. The maximum dive speed of the aircraft shall be specified.
7. Complete geometric description, including dimensioned drawings, control surfaces sizes and hinge locations, and internal arrangement of the aircraft illustrating sufficient volume for all necessary components and systems.
    a. Scaled three-views (dimensioned) and 3-D model imagery of appropriate quality are expected. The three-view must include at least:
        i. Fully dimensioned front, left, and top views
        ii. Location of aircraft aerodynamic center (from nose)
        iii. Location of average CG location (relative to nose)
        iv. Tail moment arms
    b. Diagrams and/or estimates showing that internal volume requirements are met:
        i. Cross-section showing payload layout
        ii. Systems and fuel tank layout
        iii. Fuselage centerline diagram
    c. Diagrams showing the location and functions for all aircraft systems.
8. Important aerodynamic characteristics and aerodynamic performance for key mission segments and requirements
9. Aircraft weight statement, aircraft center-of-gravity envelope reflecting payloads and fuel allocation. Establish a forward and aft center of gravity (CG) limits for safe flight.
    a. Weight assessment summary shall be shown at least at the following level of detail:
        i. Propulsion (engine/motor, batteries, controller, wiring, heat sink, cowl, strut, propeller, spinner etc. as applicable)
        ii. Airframe Structure
            1. Wing
            2. Empennage
            3. Landing Gear (including wheels tires and brakes)
            4. Fuselage
        iii. Control system (flight controls linkages, hydraulics, wires, actuators bellcranks, engine controls etc.)
        iv. Payloads
        v. Systems

1. Instruments and Avionics
2. Fuel/oil
3. Hydraulic/pneumatic/electrical systems (if chosen)

10. Propulsion system description and characterization including performance, dimensions, and weights. The selection of the propulsion system(s), sizing, and airframe integration must be supported by analysis, trade studies, and discussion
11. Summary of basic stability and control characteristics; this should include, but is not limited to static margin, pitch, roll and yaw derivatives.
12. Summary of cost estimate and a business case analysis. This assessment should identify the cost groups and drivers, assumptions, and design choices aimed at the minimization of production costs.
    a. Estimate the non-recurring development costs of the airplane including engineering, FAA certification, production tooling, facilities and labor
    b. Estimate the aircraft final cost
    c. Estimate of direct operating cost per airplane flight hour
        i. Fuel, oil, tires, brakes, battery, and other consumable quantities
        ii. Estimate of maintenance cost per flight hour
        iii. Flight crew (remote pilot) costs per hour

The design report will include trade documentation on the two major aspects of the design development, a) the concept selection trades, and b), the concept development trade studies.
The student(s) is (are) to develop and present the alternative concepts considered leading to the down-select of their preferred concept. The methods and rationale used for the down-select shall be presented. At a minimum a qualitative assessment of strengths and weaknesses of the alternatives shall be given, discussing merits, leading to a justification as to why the preferred concept was the best design. Quantitative justification of why the selected concept is the best at meeting the measures of merit(s) will strengthen the report.
In addition, the submittal shall include the major trade studies conducted justifying the optimization, sizing, architectural arrangement and integration of the specifically selected concept. Quantitative data shall be presented showing why their concept 'works' and is the preferred design compromise that best achieves the design requirements.

Specific analysis and trade studies of interest include:
Mission performance and sizing for the definition of a mission profiles.
Overall aircraft concept selection (airframe and propulsion system) vs. design requirements objectives

All concept and technology assumptions must be reasonable and justified for the EIS year.

Reference Material

https://www.altiuas.com/

https://martinuav.com/v-bat/

https://www.boeing.com/defense/autonomous-systems/hale/index.page

https://en.wikipedia.org/wiki/Northrop_Grumman_MQ-4C_Triton#:~:text=The%20Northrop%20Grumman%20MQ-4C%20Triton%20is%20an%20American,%2C%20it%20is%20an%20unmanned%20aircraft%20system%20%28UAS%29.

FAA 14 CFR Part 23
https://www.ecfr.gov/cgi-bin/text-idx?SID=4568c06ff7014def71c5330ae0a8df97&mc=true&tpl=/ecfrbrowse/Title14/14cfr23_main_02.tpl

FAA 14 CFR Part 107
https://www.ecfr.gov/cgi-bin/text-idx?SID=d5fab56e48c820847e3c1105fa7b6ab0&mc=true&node=pt14.2.107&rgn=div5

Regulatory reference material
https://www.faa.gov/uas/
https://www.faa.gov/uas/advanced_operations/section_333/

Aircraft Design References
https://en.wikipedia.org/wiki/Aircraft_design_process
http://www.darcorp.com/Books/
https://arc.aiaa.org/doi/book/10.2514/4.869112
http://www.dept.aoe.vt.edu/~mason/Mason_f/ConfigAero.html
http://www.dept.aoe.vt.edu/%7Emason/Mason_f/MRsoft.html